Program : **Diploma in Tool & Die Engineering**	
Course Code : **5118**	Course Title: **Press Tool Lab**
Semester : **5**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

CourseObjectives:

- To familiarize students with the press tool making principles and operations in Tool & Die engineering.
- To impart skills on various precision measuring instruments for taking dimensions.
- To give knowledge on theassembly & maintenance of simple stamping dies.

Course Prerequisites:

Topic	Course code	Course name	Semester
Design concepts of Press tools		Press Tool Technology	4
Limits, Fits, G&D Tolerance, Surface Finish symbols and values		Tool Engineering drawing	3
Heat treatment processes & various tool steels		Material science & Metrology	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate the press tool components of Simple Guide plate tool/progressive tool.	15	Applying
CO2	Practice the press tool assembly & maintenance of simple dies.	12	Applying
CO3	Describe the different types of presses and their applications.	6	Understanding

CO4	Demonstrate metal stamping on suitable press and Product Trial & Error runs.	9	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3					3	3
CO2	3	3					
CO3	2						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Demonstrate the press tool components of Simple Guide plate tool/progressive tool**		
M1.01	Produce the guide plate tool components –Die plate, Punch holder plate, Punch plate, Stripper plate, Top & Bottom plate, Punches, Guide pin, Guide pillars etc.	15	Applying
CO2	**Practice the press tool assembly & maintenance of simple dies.**		
M2.01	Demonstrate bench work assembly of the simple stamping dies.	6	Applying
M2.02	Employ the acceptance criteria and do the necessary rectifications.	3	Applying
M2.03	Practice on assembly & maintenance of various bending & forming tools.	3	Applying
	Lab Exam – I	1.5	
CO3	**Describe the different types of presses and their applications.**		
M3.01	Describe the type & applications of the presses according to their construction. Fly press, Arbor press, Foot press, Power press, Draw, Friction disc, Crank, Eccentric press etc.	6	Understanding

CO4	**Demonstrate metal stamping on suitable press and Product Trial & Error runs.**		
M4.01	Apply Product Trial runs on the appropriate press.	6	Applying
M 4.02	Employ the acceptance criteria and do the necessary rectifications.	3	Applying
	Lab Exam– II	1.5	

Text/ Reference:

T/R	Book Title/Author
T1	J.R.Paquin, Die design fundamentals, Industrial Press Inc, 1990.
R1	Donaldson, Tool Design, Tata McGraw-hill Book company, 23rd edition, 2006
R2	Donald F. Eary., Edward A. Reed, Techniques of Press working sheet metal, Prentice-Hall,Inc. Second Edition, 1974.

Online Resources:

Sl.No	Website Link
1	https://www.researchgate.net/publication/264982108_Types_of_press_tools
2	http://www.ijirset.com/upload/2016/may/285_Process.pdf
3	http://staff.uny.ac.id/sites/default/files/pendidikan/aan-ardian-mpd/1g-handbook-die-design-2nd-edition.pdf
4	http://ijcttjournal.org/Volume4/issue-7/IJCTT-V4I7P142.pdf